Mathematical methods for scientific computing
Vector Calculus

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Mathematical methods for scientific computing

- Arguably, scientific computing could be considered the field of turning equations into code, to solve problems which are otherwise intractable
- We start with a system of equations, e.g.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$$

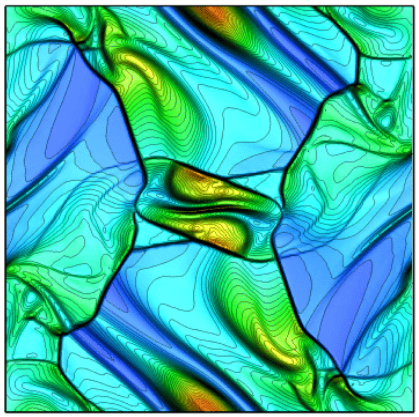
$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + \mathbf{I} p) = 0$$

$$\frac{\partial E}{\partial t} + \nabla \cdot [(E + p)\,\mathbf{v}] = 0$$

- From these, numerical methods for their solution are developed

$$\mathbf{u}_i^{n+1} = \mathbf{u}_i^n - \frac{\Delta t}{\Delta x}\left(\mathbf{f}_{i+1/2}^{\mathrm{FORCE}} - \mathbf{f}_{i-1/2}^{\mathrm{FORCE}}\right)$$

- And finally, we get to simulation results



- In order to derive, and understand the properties of these methods, we need appropriate mathematical techniques

Mathematical methods for scientific computing

- Throughout the MPhil in scientific computing, you will frequently need to use the mathematical techniques in this course, either for deriving or manipulating the equations you are going to solve, or for processing and understanding solution data
- Some of this course is likely to have been covered in previous courses, but that may have been a while ago, or approached in a different way
- In this course, the aim is to set the methods lectured in the context of numerical methods and scientific computing
- Accompanying this course are a series of problems - these will help reinforce the content of this course, but will likely take longer than the gap between lectures, so should can be completed throughout term, and for revision

Assumed knowledge

- Whilst this course will build on your previous maths background, it's worth repeating a few of the key expressions which will be used repeatedly, just in case they've faded from memory a little

1. The first fundamental theorem of calculus

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\int_{x_0}^{x} f(x')\mathrm{d}x'\right) = f(x)$$

(providing f is continuous)

2. The second fundamental theorem of calculus

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\int_{x_0}^{x_1} f(x)\mathrm{d}x\right) = \left(\int_{x_0}^{x_1} \frac{\mathrm{d}f}{\mathrm{d}x}\mathrm{d}x\right) = f(x_1) - f(x_0)$$

(providing f is differentiable)

Assumed knowledge

3. Taylor series

$$f(x) = f(a) + \left.\frac{\partial f}{\partial x}\right|_{x=a}(x-a) + \left.\frac{1}{2}\frac{\partial^2 f}{\partial x^2}\right|_{x=a}(x-a)^2 + \left.\frac{1}{3!}\frac{\partial^3 f}{\partial x^3}\right|_{x=a}(x-a)^3 + \dots$$

or

$$f(x+a) = f(x) + a\,\frac{\partial f}{\partial x} + a^2\,\frac{1}{2}\frac{\partial^2 f}{\partial x^2} + a^3\,\frac{1}{3!}\frac{\partial^3 f}{\partial x^3} + \dots$$

4. Fourier series for a function with period L

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty}\left(a_n \cos\left(\frac{2\pi n x}{L}\right) + b_n \sin\left(\frac{2\pi n x}{L}\right)\right)$$

$$a_n = \frac{2}{L}\int_{x_0}^{x_0+L} f(x)\cos\left(\frac{2\pi n x}{L}\right), \quad b_n = \frac{2}{L}\int_{x_0}^{x_0+L} f(x)\sin\left(\frac{2\pi n x}{L}\right)$$

Assumed knowledge

5. Binomial expansion, in particular,

$$(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots$$

6. Stokes' theorem

$$\oint_C \mathbf{F} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{F}) \cdot \mathrm{d}S$$

7. Divergence theorem

$$\int (\nabla \cdot \mathbf{F}) \mathrm{d}V = \int_S \mathbf{F} \cdot \mathrm{d}S$$

Note - despite these last two equations often being core components of vector calculus, we actually won't really use these - they appear more in analytic calculations than computational ones

Notation note

- In keeping with standard mathematical notation, vector quantities will always be emboldened, e.g. $\mathbf{a}$ is a vector, a is a scalar
- Often, vectors will be represented by lower case letters, matrices by upper case, but this is not universal
- The reason for this is conflicting conventions, for example, magnetic field is typically $\mathbf{B}$
- I shall attempt to keep matrices non-italic and non-bold (but may forget, sometimes) - context is often more important than typeface
- And when we move to index notation, indices are the only important thing to look at - they provide the context

Outline

1. Vector calculus
2. Index notation
3. Tensors

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Vector calculus and evolution equations

- The first topic we shall consider is vector calculus, with a key focus on index notation
- Hopefully, the inclusion of this topic should be expected, as when describing physical systems, we would both expect to find vectors (velocity, relative positions) and calculus (gradients and rates of change)
- One example, particularly relevant to continuum modelling, which provides a good example of why this is important is the compressible Euler equations (the Navier-Stokes equations do too, but take up a little more space)

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$$

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + \mathbf{I} p) = 0$$

$$\frac{\partial E}{\partial t} + \nabla \cdot [(E + p)\,\mathbf{v}] = 0$$

What are these equations?

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$$

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + \mathbf{I} p) = 0$$

$$\frac{\partial E}{\partial t} + \nabla \cdot [(E + p)\,\mathbf{v}] = 0$$

- Those on the continuum stream will see these equations a lot, and those on the HPC stream won't avoid them elsewhere either
- Here, they have been picked, partly because I'm very familiar with them, but also because the natural form for expressing them requires vector calculus operations
- If we want to rearrange any such equations, we will need to use vector calculus
- Physically, they are the conservation of mass (or density), momentum and energy
- And, for example, the first of these says that the change in mass over time is related to the gradient in momentum (which can be considered a rate of flow of mass)

Vector form of equations

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$$

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + \mathbf{I}p) = 0$$

$$\frac{\partial E}{\partial t} + \nabla \cdot [(E + p)\,\mathbf{v}] = 0$$

- Real-world simulations typically cannot be described by a single variable; we need to consider the interaction of multiple variables in a **coupled system**
- The density depends on the momentum, but the momentum itself also depends on density (and pressure)
- It is therefore common to find these systems described in "vector form"

$$\frac{\partial}{\partial t} \begin{pmatrix} \rho \\ \rho \mathbf{v} \\ E \end{pmatrix} + \nabla \cdot \begin{pmatrix} \rho \mathbf{v} \\ \rho \mathbf{v} \otimes \mathbf{v} + \mathbf{I}p \\ (E + p)\,\mathbf{v} \end{pmatrix} = 0$$

Unpacking this vector form

$$\frac{\partial}{\partial t}\begin{pmatrix}\rho\\ \rho\mathbf{v}\\ E\end{pmatrix}+\nabla\cdot\begin{pmatrix}\rho\mathbf{v}\\ \rho\mathbf{v}\otimes\mathbf{v}+\mathbf{I}p\\ (E+p)\,\mathbf{v}\end{pmatrix}=0$$

- Although a very concise definition of the equations, there is perhaps some unexpected notation
- We have vectors within vectors
- At this stage, we also can't write out these terms explicitly, since we haven't chosen our coordinate system yet
- This would make the second vector a matrix, so now we also have the divergence of a matrix
- This does have a reasonable mathematical definition, however, and we shall come to this shortly

Coordinate-independent expressions

$$\frac{\partial}{\partial t}\begin{pmatrix}\rho\\ \rho\mathbf{v}\\ E\end{pmatrix}+\nabla\cdot\begin{pmatrix}\rho\mathbf{v}\\ \rho\mathbf{v}\otimes\mathbf{v}+\mathbf{I}p\\ (E+p)\,\mathbf{v}\end{pmatrix}=0$$

- It is common to reduce dimensionality in problems by appealing to symmetries, for example cylindrical symmetry. The equation above holds, regardless of whether

$$\mathbf{v}=v_x\mathbf{e}_i+v_y\mathbf{e}_j+v_z\mathbf{e}_k \qquad \text{or} \qquad \mathbf{v}=v_r\mathbf{e}_r+v_\theta\mathbf{e}_\theta+v_\phi\mathbf{e}_\phi$$

- Once you know your coordinate system, you might be able to simply use e.g.

$$\begin{pmatrix}\rho\\ \rho\mathbf{v}\\ E\end{pmatrix}=\begin{pmatrix}\rho\\ \rho v_x\\ \rho v_y\\ \rho v_z\\ E\end{pmatrix}$$

- However, this independent approach avoids (or hides) an additional complication, when expressing the equation in your chosen coordinates, the basis vectors themselves might have non-zero derivatives

Vector operations - quick review

- For now, we shall assume we are working in Cartesian coordinates and define everything in terms of the three directions x, y and z.
- In these coordinates, gradient, divergence and curl are the key calculus operators

$$\nabla\alpha = \begin{pmatrix} \frac{\partial\alpha}{\partial x} \\ \frac{\partial\alpha}{\partial y} \\ \frac{\partial\alpha}{\partial z} \end{pmatrix}$$

- Gradient operates on scalar fields, and provides their rate of change in arbitrary directions as a vector field
- Note - $\nabla\alpha$ is perpendicular to the field, α

Vector operations - quick review

- Divergence, or just div, can be considered the rate at which a vector field enters or leaves a given volume

$$\nabla \cdot \mathbf{F} = \lim_{V \to 0} \frac{1}{V} \int_S \mathbf{F} \cdot \hat{\mathbf{n}} \mathrm{d}S$$

$$\nabla \cdot \mathbf{a} = \frac{\partial a_x}{\partial x} + \frac{\partial a_y}{\partial y} + \frac{\partial a_z}{\partial z}$$

- This is why it appears in expressions like conservation of mass - it allows us to track mass loss/gain in a volume
- Mathematically, this is not technically a dot product, but it is nearly universally used this way

Vector operations - quick review

- Curl is a measure of circulation - related to the **rate of rotation**, not directly to the rotation itself
- Also has a limiting definition

$$\nabla \times \mathbf{F} = \lim_{V \to 0} \frac{1}{V} \int_S \hat{\mathbf{n}} \times \mathbf{F} \mathrm{d}S$$

$$\nabla \times \mathbf{a} = \begin{vmatrix} \mathbf{e}_i & \mathbf{e}_j & \mathbf{e}_k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_x & a_y & a_z \end{vmatrix} = \begin{pmatrix} \frac{\partial a_z}{\partial y} - \frac{\partial a_y}{\partial z} \\ \frac{\partial a_x}{\partial z} - \frac{\partial a_z}{\partial x} \\ \frac{\partial a_y}{\partial x} - \frac{\partial a_x}{\partial y} \end{pmatrix}$$

- Commonly encountered when dealing with electric and magnetic fields, since these act perpendicularly to each other, e.g.

$$\frac{\partial \mathbf{B}}{\partial t} + \nabla \times \mathbf{E} = 0$$

- Again, technically this is not a true cross product mathematically

More vector operations

- Sometimes less familiar operations may appear
- For example, the outer product, $\otimes$, and the divergence of a matrix (tensor)

$$\mathbf{a} \otimes \mathbf{b} = \begin{pmatrix} a_x b_x & a_x b_y & a_x b_z \\ a_y b_x & a_y b_y & a_y b_z \\ a_z b_x & a_z b_y & a_z b_z \end{pmatrix}$$

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + \mathbf{I} p) = 0$$

- The physical use of this term is, for example, when a quantity moving in one direction is affected by quantities in other directions
- For example, a gradient in ρv_x can exist in the y-direction, and thus $x-$momentum can move in the $y-$direction
- Some papers are lazy, and write this as $\rho \mathbf{v}\mathbf{v}$ (since it is the non-dot product multiplication of two vectors, column $\times$ row)

More vector operations

- The divergence of a matrix is another term which needs consideration, and clarification

$$\nabla \cdot \mathrm{A} = \begin{pmatrix} \frac{\partial A_{xx}}{\partial x} + \frac{\partial A_{xy}}{\partial y} + \frac{\partial A_{xz}}{\partial z} \\ \frac{\partial A_{yx}}{\partial x} + \frac{\partial A_{yy}}{\partial y} + \frac{\partial A_{yz}}{\partial z} \\ \frac{\partial A_{zx}}{\partial x} + \frac{\partial A_{zy}}{\partial y} + \frac{\partial A_{zz}}{\partial z} \end{pmatrix}$$

- At this point, you may ask why is $\nabla \cdot \mathrm{A}$ defined like this? Why not

$$\nabla \cdot \mathrm{A} = \begin{pmatrix} \frac{\partial A_{xx}}{\partial x} + \frac{\partial A_{yx}}{\partial y} + \frac{\partial A_{zx}}{\partial z} \\ \frac{\partial A_{xy}}{\partial x} + \frac{\partial A_{yy}}{\partial y} + \frac{\partial A_{zy}}{\partial z} \\ \frac{\partial A_{xz}}{\partial x} + \frac{\partial A_{yz}}{\partial y} + \frac{\partial A_{zz}}{\partial z} \end{pmatrix} ?$$

- Unfortunately, the answer is that it could be defined this way. You are often expected to know which version you use
- Sometimes, the first of these is divA, and the second $\nabla \cdot \mathrm{A}$
- Alternatively, a less vague way of defining these equations could be used...

Outline

1. Vector calculus
2. Index notation
3. Tensors

Index notation

- Some of these quantities are not natural to work with, for example, we may want to write $\nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v})$ as partial derivatives of just $\mathbf{v}$
- Additionally, for some cases, such as elasticity, we end up with rank-4 tensors (like a matrix, but in four dimensions...)
- In such cases, **index notation** makes life much more straightforward
- For example,

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + \mathbf{I}p) = \nabla \cdot \boldsymbol{\tau} + \rho \mathbf{f}$$

(where $\boldsymbol{\tau}$ is a matrix) becomes

$$\frac{\partial \rho v_i}{\partial t} + \nabla_j \left(\rho v_i v_j + \delta_{ij} p\right) = \nabla_j \tau_{ij} + \rho f_i$$

- As for why this makes life straightforward, please bear with me...

Reading index notation

$$\frac{\partial \rho v_i}{\partial t} + \nabla_j \left(\rho v_i v_j + \delta_{ij} p\right) = \nabla_j \tau_{ij} + \rho f_i$$

- The (unique) indices tell you what you are dealing with
- Scalars have no indices, e.g. ρ and p
- Vectors have one index, e.g.

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \quad \rightarrow \quad v_i \mathbf{e}_i \quad \text{or} \quad v_i$$

- In Cartesian coordinates, the $(1, 2, 3)$ and (x, y, z) subscripts are often used to mean the same thing $(v_1, v_2, v_3)^T = (v_x, v_y, v_z)^T$
- Note - $v_i \mathbf{e}_i$ could be considered partial index notation - vector components are represented as indices, but operators (e.g. dot products) aren't

Reading index notation

$$\frac{\partial \rho v_i}{\partial t} + \nabla_j \left(\rho v_i v_j + \delta_{ij} p\right) = \nabla_j \tau_{ij} + \rho f_i$$

- Matrices have two indices, the first gives the row, and the second gives the column

$$\boldsymbol{\tau} = \begin{pmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{pmatrix} = \begin{pmatrix} \tau_{11} & \tau_{12} & \tau_{13} \\ \tau_{21} & \tau_{22} & \tau_{23} \\ \tau_{31} & \tau_{32} & \tau_{33} \end{pmatrix} \quad \rightarrow \quad \tau_{ij}$$

$$\mathrm{I} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \delta_{11} & \delta_{12} & \delta_{13} \\ \delta_{21} & \delta_{22} & \delta_{23} \\ \delta_{31} & \delta_{32} & \delta_{33} \end{pmatrix} \quad \rightarrow \quad \delta_{ij}$$

- When written in index notation, the identity matrix is typically called the **Kronecker delta**

Combined quantities

$$\frac{\partial \rho v_i}{\partial t} + \nabla_j \left(\rho v_i v_j + \delta_{ij} p\right) = \nabla_j \tau_{ij} + \rho f_i$$

- As might be expected, everything separated by addition, subtraction or an equals sign must be the same **rank**, i.e. the same type of thing
- Probably an obvious statement (you can't add a vector to a matrix) but helpful when checking your calculations
- Similarly, everything within a set of parentheses must be the same thing
- So here, both $\rho v_i v_j$ and $\delta_{ij} p$ have two indices and both are matrices
- $\delta_{ij} p$ is a scalar multiplied by a matrix; we clearly expect this to be a matrix
- As is the outer product quantity, $\rho v_i v_j$, it is now very clear to see what the entries of this matrix should be

Free indices

$$\frac{\partial \rho v_i}{\partial t} + \nabla_j \left(\rho v_i v_j + \delta_{ij} p\right) = \nabla_j \tau_{ij} + \rho f_i$$

- We have so far avoided talking about what the $\nabla_j \tau_{ij}$ term is
- It should be a vector, since $\frac{\partial \rho v_i}{\partial t}$ and ρf_i are vectors
- It also has one index repeated (j) and one which only appears once (i)
- A free index is one that appears **only once** per term
- This is what determines what type of quantity you have
- Repeated indices represent a summation

Repeated indices

- Repeated indices represent a sum over that index - **Einstein summation convention**

$$v_i v_i = \sum_{i=1}^{3} v_i v_i = v_1 v_1 + v_2 v_2 + v_3 v_3 = \mathbf{v} \cdot \mathbf{v} = v^2$$

- Used for simplifying notation
- This index can be repeated in two different quantities, such as the dot product above
- Or a repeated index within a single quantity, as in Hooke's law for stress and strain

$$\sigma_{ij} = \lambda \delta_{ij} e_{kk} + 2\mu e_{ij}$$

- Here,

$$e_{kk} = e_{11} + e_{22} + e_{33} = \mathrm{Tr}(\mathbf{e})$$

Dummy indices

- For free indices, the index letter must be used consistently, for example, these are all different quantities

$$C_{ij} = A_{ij} + B_{ij} \quad \hat{C}_{ij} = A_{ji} + B_{ji}, \quad \bar{C} = A_{ij} + B_{ji}$$

- Repeated indices are sometimes known as dummy indices, because they can be freely renamed whenever you like

$$v_i v_i = \sum_{i=1}^{3} v_i v_i = v_1 v_1 + v_2 v_2 + v_3 v_3 = \sum_{j=1}^{3} v_j v_j = v_j v_j$$

- As a result, we frequently rename repeated indices to make calculations easier.

Vector and matrix multiplication

- We have sort of introduced this with the dot product and the summation convention
- But, unsurprisingly, since vector-matrix and matrix-matrix multiplication involves summation over rows or columns, these can be described using index notation
- For example, since

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} a_{11}b_1 + a_{12}b_2 \\ a_{21}b_1 + a_{22}b_2 \end{pmatrix} = \begin{pmatrix} \sum a_{1j}b_j \\ \sum a_{2j}b_j \end{pmatrix}$$

  we can easily see this can be written

$$\mathbf{Ab} \quad \rightarrow \quad A_{ij}b_j$$

- Similarly,

$$\mathrm{AB} \quad \rightarrow \quad A_{ij}B_{jk}$$

Indices - summary

- v_i, ϵ_{iij}, $a_{ij}b_j$: Vector (one free index)
- τ_{ij}, $v_i v_j$, $a_{ij}b_{jk}$: Matrix (two free indices)
- τ_{ii}: Scalar (no free indices)
- τ_{11}: A specific entry of the matrix τ_{ij}, $i = 1$, $j = 1$
- C_{ijkl}: Rank four object (this symbol is used for the elasticity tensor)
- C_{iikl}: Matrix (two free indices)
- C_{iiil}, ϵ_{iii}, $a_i + B_{ij}$: ??? Something has gone wrong

Back to grad, div and curl

- The grad operator, ∇, is a vector operator, so can be written with index notation

$$\nabla = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \rightarrow \nabla_i = \frac{\partial}{\partial x_i}$$

- Grad and div are now straightforward,

$$\nabla \alpha \rightarrow \nabla_i \alpha, \qquad \nabla \cdot \mathbf{a} \rightarrow \nabla_i a_i$$

- Curl requires the addition of a new quantity, the Levi-Civita symbol

$$\nabla \times \mathbf{a} \rightarrow \epsilon_{ijk} \nabla_j a_k, \qquad \epsilon_{ijk} = \begin{cases} +1 & \text{if } (i,j,k) \text{ is } (1,2,3), (2,3,1), \text{ or } (3,1,2), \\ -1 & \text{if } (i,j,k) \text{ is } (3,2,1), (1,3,2), \text{ or } (2,1,3), \\ 0 & \text{if } i=j, \text{ or } j=k, \text{ or } k=i \end{cases}$$

Levi-Civita symbol

$$\nabla \times \mathbf{a} = \epsilon_{ijk} \nabla_j a_k$$

- There is one free index in $\epsilon_{ijk} \nabla_j a_k$, so as expected, this is a vector term
- The term $\nabla_j a_k$ is a matrix and the Levi-Civita symbol a '3-D matrix'

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\nabla_j a_k = \begin{pmatrix} \frac{\partial a_x}{\partial x} & \frac{\partial a_y}{\partial x} & \frac{\partial a_z}{\partial x} \\ \frac{\partial a_x}{\partial y} & \frac{\partial a_y}{\partial y} & \frac{\partial a_z}{\partial y} \\ \frac{\partial a_x}{\partial z} & \frac{\partial a_y}{\partial z} & \frac{\partial a_z}{\partial z} \end{pmatrix}$$

- The quantity $\epsilon_{ijk} \nabla_j a_k$ then multiplies each 'slice' of ϵ_{ijk} with $\nabla_j a_k$
- Also note that any cross product is written in index notation with the Levi-Civita symbol,

$$\mathbf{a} \times \mathbf{b} = \epsilon_{ijk} a_j b_k$$

Vector calculus in index notation

- Working in index notation has a few advantages
- Firstly, notation for matrices, and 'higher dimensional matrices' is straightforward, since there is no universal convention for these quantities
- Simplification of expressions is also more straightforward - there is no need to memorise any relationships
- Well... maybe one relationship

$$\epsilon_{ijk}\epsilon_{ilm} = \delta_{jl}\delta_{km} - \delta_{jm}\delta_{kl}$$

- Additionally, dealing with derivatives is a lot easier, as in index notation, you are treating things at the level of scalar quantities

Changing indices with the Kronecker delta

- One of the key features of vector calculus operations in index notation is the use of the Kronecker delta to change indices
- Sometimes this is obvious

$$\delta_{ij} a_j = a_i$$

- This is really just saying that multiplying a vector by the identity matrix does not change the vector - which is the definition of the identity matrix!
- Sometimes it takes a bit more thought to work out what is happening

$$\delta_{ij} A_{ij} = A_{ii}, \quad \delta_{im}\delta_{jn} B_{mnkl} = B_{ijkl}$$

- And occasionally, we work backwards

$$\nabla_j \rho v_i v_j + \nabla_i p = \nabla_j \left(\rho v_i v_j + \delta_{ij} p\right)$$

Some examples

- All 'classic' vector calculus identities can be demonstrated using index notation

$$\nabla \cdot (\mathbf{a} \times \mathbf{b}) = (\nabla \times \mathbf{a}) \cdot \mathbf{b} - \mathbf{a} \cdot (\nabla \times \mathbf{b})$$

- Throughout the continuum courses, we will make use of index notation for expressing equations in 'desired' forms
- If we want to solve the Maxwell-Faraday equation (with a few other coupled equations, and making use of Ohm's law),

$$\frac{\partial \mathbf{B}}{\partial t} + \nabla \times \mathbf{E} = 0, \qquad \eta \mathbf{J} = \mathbf{E} + \mathbf{v} \times \mathbf{B}, \qquad \eta = 0$$

- Then it is often desirable to express all spatial derivatives through a divergence operator

$$\frac{\partial \mathbf{B}}{\partial t} + \nabla \cdot (\mathbf{B} \otimes \mathbf{v} - \mathbf{v} \otimes \mathbf{B}) = 0$$

A disclaimer

- Technically, index notation does not just use subscripts, but also superscripts

$$a^i = \begin{pmatrix} a^x \\ a^y \\ a^z \end{pmatrix}, \qquad a_i = (a_x, a_y, a_z)$$

- Because we tend to focus on Cartesian coordinates, we ignore this description (we have $a^x = a_x$ and so on)
- But, in more general coordinate systems, **this is not true**, and there is a transformation required to move between the two forms
- We rarely encounter this in this MPhil; we even write non-Cartesian coordinates in a 'Cartesian-like' manner
- Fusion plasma equilibrium literature is a notable exception - non-orthogonal coordinate systems are common

Outline

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2. Index notation
3. Tensors

Tensors

- Using index notation makes it easy to write down things like a 'three-dimensional matrix' in a straightforward way
- And when dealing with these sort of quantities, it is common to refer to them as tensors
- Though, in reality, tensors have a slightly stricter definition
- This starts with vectors, which are expressed in a given coordinate system through **basis vectors**

$$\mathbf{a} = a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + a_3\mathbf{e}_3$$
$$\mathbf{a} = a'_1\mathbf{e}'_1 + a'_2\mathbf{e}'_2 + a'_3\mathbf{e}'_3$$

- For example, we may have Cartesian basis vectors, $\mathbf{e}_x$, $\mathbf{e}_y$, $\mathbf{e}_z$ (often written $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$), or spherical polar basis vectors, $\mathbf{e}_r$, $\mathbf{e}_\phi$, $\mathbf{e}_\theta$
- Coordinates transform through

$$a_i = \frac{\partial x_i}{\partial x'_j} a'_j \quad \text{or} \quad a_i = R_{ij}a'_j \quad \text{or} \quad a_i = L_{ij}a'_j$$

Why do we care about tensors?

- Vectors can always be transformed through the Jacobian
- A tensor is a generalisation of this - any quantity that also transforms according to these transformation rules, e.g.

$$t_{ij} = R_{ik} R_{jl} t'_{kl}$$

or

$$t_{ijkl} = R_{im} R_{jn} R_{kp} R_{lq} t'_{mnpq}$$

- The number of indices is also known as its **rank**
- One of the key aspects of a tensor is that the quantity represents the same thing, **no matter what coordinate system we are working in** (though individual entries will, of course, be different)
- This is important if you want consistent definitions of quantities such as stress, or elastic deformation
- We see this importance in **invariant quantities**

Invariant quantities

- Invariant quantities are those which are the same, no matter the coordinate system
- For a rank-2 tensor, the invariant quantities come from finding its eigenvalues

$$\begin{vmatrix} a_{11}-\lambda & a_{12} & a_{13} \\ a_{21} & a_{22}-\lambda & a_{23} \\ a_{31} & a_{32} & a_{33}-\lambda \end{vmatrix} = 0$$

- This determinant can be written

$$\lambda^3 - I_1\lambda^2 + I_2\lambda - I_3 = 0$$

$$I_1 = a_{ii} = a_{11} + a_{22} + a_{33} = \mathrm{tr}(\mathrm{A})$$

$$I_2 = \frac{1}{2}\left(a_{ii}a_{jj} - a_{ij}a_{ji}\right) = \frac{1}{2}\left[(\mathrm{tr}(\mathrm{A}))^2 - \mathrm{tr}(\mathrm{A}^2)\right]$$

$$I_3 = \det|\mathrm{A}|$$

- I_1, I_2 and I_3 are the **invariants** of a rank-2 tensor in 3D

Invariants, pressure and stress

- When modelling any continuum material, stress (σ_{ij}) is a rank-2 tensor which describes how the material is being pushed (or pulled) in any direction
- Pressure (p) is a scalar quantity, which gives some information about how much stress the material is under
- Perhaps unsurprisingly, the pressure experienced at any point within a material does not depend on the coordinate system used to calculate it - it is an invariant quantity
- In fact,

$$p = \frac{1}{3} I_1 = \frac{1}{3} \mathrm{tr}(\sigma)$$

Matrices which aren't tensors

- There are two commonly encountered quantities which are matrices, but not tensors
- The first is the coordinate transformation, R_{ij},

$$\mathbf{e} = \frac{\partial \mathbf{x}}{\partial \mathbf{x}'}\mathbf{e}' \quad \text{or} \quad e_i = R_{ij}e'_j$$

- The second is the **deformation gradient**, F_{ij}

$$F_{ij} = \frac{\mathrm{d}x_i}{\mathrm{d}X_i}$$

- This transforms an infinitesimal particle from its reference configuration $\mathrm{d}\mathbf{X}$ to its current position $\mathrm{d}\mathbf{x}$, by deformation alone (no rotation or translation)

What are these matrices

- Both these examples take one vector ($\mathbf{e}'$ or $\mathbf{X}$), and transform them to a new vector ($\mathbf{e}$ or $\mathbf{x}$)
- These quantities **transform as vectors**, i.e. would need one coordinate transform, whilst a regular rank-2 tensor would require two
- On inspection, perhaps not surprising - one of the indices is already in the coordinate system we are transforming to
- Such quantities are known as **two-point tensors** or **double vectors**
- This name does not always make it to the literature though...